

2017 普物 820

一、

$$1. a = \frac{v^2}{\rho} = g \Rightarrow \rho = \frac{v^2}{g} = \frac{v^2 \cos^2 \theta}{g}$$

$$2. \vec{a} = \frac{\vec{F}}{m} = \frac{4t^3}{m} \hat{i} \quad x = \frac{1}{2}at^2 = 2t^3 \hat{i} \quad y = 2t \hat{j}$$

$$\vec{r} = 2t^3 \hat{i} + 2t \hat{j}$$

$$3. F = -(k_1 + k_2)x \quad \frac{d^2x}{dt^2} + \frac{k_1 + k_2}{m}x = 0 \quad \varphi_0 = 0$$

$$x = x_0 \cos \sqrt{\frac{k_1 + k_2}{m}} t$$

$$4. l = l_0 \sqrt{1 - \frac{v^2}{c^2}} \Rightarrow v = c \sqrt{1 - \frac{l^2}{l_0^2}} \quad p = mv = \frac{m_0 l_0}{l} \sqrt{1 - \frac{l^2}{l_0^2}}$$

$$5. f_1 = \frac{330}{330-2} \cdot 400 \approx 402.44 \text{ Hz} \quad f_2 = \frac{330}{330+2} \cdot 400 = 397.59 \text{ Hz}$$

$$\Delta f = 4.85 \text{ Hz}$$

$$6. T_2 = 2T_1 \quad \lambda = \frac{kT}{\sqrt{2\pi} \rho} \quad \therefore \frac{\lambda_2}{\lambda_1} = 2$$

$$7. U_{\text{max}} = \frac{q}{4\pi\epsilon_0 R} = \frac{qR}{4\pi\epsilon_0 R^2} = RE = 3 \times 10^6 \text{ V}$$

$$8. B = \frac{\mu_0 I}{2\pi a} \cdot 2 = \frac{\mu_0 I}{\pi a}$$

$$9. dI = \frac{2dq}{r} = \frac{2w}{2\pi} dq = \frac{w}{\pi} \frac{q}{l} dr \quad dP_m = \pi r^2 dI = \frac{wq}{l} r dr$$

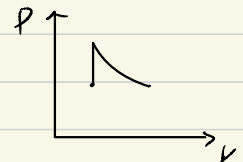
$$P_m = \int_0^{\frac{l}{2}} \frac{wq}{l} r dr = \frac{1}{24} wq l^2$$

$$10. 2\pi k \quad \frac{1}{4}$$

$$11. h\nu = A + E_k \Rightarrow A = h\nu - E_k$$

$$2h\nu = A + eU \Rightarrow U = \frac{2h\nu - h\nu + E_k}{e} = \frac{h\nu + E_k}{e}$$

$$12. E_k = h\frac{c}{\lambda} - h\frac{c}{\lambda}$$



二、

$$\begin{cases} mg - T_1 = ma_1 \\ T_2 - mg = ma_2 \\ T_1 \cdot 2r - T_2 \cdot r = J\beta \\ a_1 = 2pr = 2a_2 \\ J = \frac{9}{4} mr^2 \end{cases} \Rightarrow \begin{cases} 2T_1 - mg = \frac{11}{5} ma_2 \\ 2mg - 2T_1 = 4ma_2 \end{cases}$$

$$\Rightarrow a_2 = \frac{2}{19}g, \beta = \frac{2g}{19r}$$

三、

$$\begin{cases} m v_0 \frac{l}{2} = m \frac{v_0}{2} \frac{l}{2} + \frac{1}{3} M L^2 \omega \\ \frac{1}{2} \frac{1}{3} M L^2 \omega = Mg \frac{l}{2} \end{cases} \Rightarrow v_0 = \frac{4}{3} \frac{M}{m} \sqrt{3gl}$$

四、

$$(1) \omega = \frac{2\pi}{T} = 4\pi \text{ rad/s} \quad k = \frac{2\pi}{\lambda} = \frac{\pi}{2} \quad \varphi_0 = 0$$

$$y = 0.1 \cos(4\pi t - \frac{\pi}{2}x)$$

$$(2) t = \frac{T}{4} = \frac{1}{8} \text{ s}, \quad x = \frac{\lambda}{4} = 2.5 \text{ m} \quad y = 0.1 \text{ m}$$

$$(3) t = \frac{T}{2} = \frac{1}{4} \text{ s}, \quad x = \frac{\lambda}{4} = 2.5 \text{ m} \quad v = \frac{dy}{dt} = -\frac{1}{10} \cdot 4\pi \sin(4\pi t - \frac{\pi}{2}x) = -\frac{2}{5}\pi \sin(4\pi t - \frac{\pi}{2}x)$$

$$v = -\frac{2}{5}\pi \text{ m/s} \quad \text{方向向y轴负方向}$$

五、

$$(1) Q = \Delta E + A \quad A = \frac{1}{2}(p - p_0)(V - V_0) = \frac{3}{2}p_0V_0$$

$$\Delta E = C_M \Delta T = \frac{5}{2}R(T - T_0) = \frac{5}{2}R \cdot 7T_0 = \frac{35}{2}RT_0 = \frac{35}{2}p_0V_0$$

$$\therefore Q_I = 19p_0V_0 = 1919 \text{ J}$$

$$(2) C_{m,n} = C_{v,n} - \frac{R}{n-1} \quad C_{\frac{1}{2},n} = \frac{5}{2}R - \frac{R}{\frac{1}{2}-1} = \frac{9}{2}R \quad p_0V_0^{\frac{1}{2}} = 4p_0(2V_0)^{\frac{1}{2}}$$

$$\therefore V_3 = 32V_0 \quad \text{即 } T_3 = 32T_0 \quad \therefore Q_{II} = C_{m,n} \Delta T = \frac{9}{2}R(32T_0 - 8T_0) = 108RT_0 = 108p_0V_0$$

$$\therefore Q_{\Sigma} = (108 + 19)p_0V_0 = 12821 \text{ J}$$

六、

$$(1) D_1 = \epsilon_0 \epsilon_{r1} E = \epsilon_0 \epsilon_{r1} \frac{U}{d} \quad D_2 = \epsilon_0 \epsilon_{r2} \frac{U}{d}$$

$$(2) E_1 = E_2 = \frac{U}{d}$$

$$(3) P_1 = \epsilon_0(\epsilon_{r1}-1) \frac{U^2}{d} \quad P_2 = \epsilon_0(\epsilon_{r2}-1) \frac{U^2}{d}$$

$$(4) C = C_1 + C_2 = \frac{\epsilon_0 \epsilon}{2d} (\epsilon_{r1} + \epsilon_{r2})$$

七、

$$\psi = N\Phi = N \cdot \iint \vec{B} \cdot d\vec{s} = N \cdot \int_{R_1}^{R_2} \frac{\mu_0 i}{2\pi r} b dr = \frac{\mu_0 N b i}{2\pi} \ln \frac{R_2}{R_1}$$

$$\therefore M = \frac{\psi}{i} = \frac{\mu_0 N b}{2\pi} \ln \frac{R_2}{R_1}$$

$$\therefore \Phi' = MI = \frac{\mu_0 N^2 b i_0}{2\pi} \ln \frac{R_2}{R_1} \sin \omega t$$

$$\therefore E = -\frac{d\Phi'}{dt} = -\frac{\mu_0 N^2 b i_0 \omega}{2\pi} \ln \frac{R_2}{R_1} \cos \omega t$$

八、

$$d \sin \theta = \frac{1}{2}d = j\lambda_1 = 2\lambda \quad \Rightarrow d = 4\lambda_1 = 2400 \text{ nm}$$

$$p = \frac{\lambda}{\Delta \lambda_1} = \frac{600}{3400} = jN = 2N \quad \Rightarrow N = 10^6 \text{ 个}$$

$$\text{第三级缺级} \quad \Rightarrow a = \frac{b}{j} = \frac{k}{5}d \quad (k=1, 4)$$

$$\therefore a = 800 \text{ nm 或 } 1600 \text{ nm}$$